Fiscal Year:	FY 2005	**Task Last Updated:**	FY 01/22/2005
PI Name:	You, Li Ph.D.		
Project Title:	Theoretical studies of the generation, detection, and applications of massively entangled states of Bose condensed atoms		
Division Name:	Physical Sciences		
Program/Discipline:	FUNDAMENTAL PHYSICS		
Program/Discipline--Element/Subdiscipline:	FUNDAMENTAL PHYSICS--Laser cooled atomic physics		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
PI Email:	Li.You@Physics.gatech.edu	**Fax:**	FY 404-894-9958
PI Organization Type:	UNIVERSITY	**Phone:**	404-894-2107
Organization Name:	Georgia Institute of Technology		
PI Address 1:	School of Physics, Mail Code 0430		
PI Address 2:	837 State Street		
PI Web Page:			
City:	Atlanta	**State:**	GA
Zip Code:	30332-0430	**Congressional District:**	5
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	2002 Physical Sciences 02-OBPR-03
Start Date:	04/01/2004	**End Date:**	09/30/2006
No. of Post Docs:	0	**No. of PhD Degrees:**	
No. of PhD Candidates:	0	**No. of Master' Degrees:**	
No. of Master's Candidates:	1	**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:	0	**Monitoring Center:**	NASA JPL
Contact Monitor:	Maleki, Lute	**Contact Phone:**	818-354-3688
Contact Email:	lute.maleki@jpl.nasa.gov		
Flight Program:			
Flight Assignment:			
Key Personnel Changes/Previous PI:			
COI Name (Institution):			
Grant/Contract No.:			
Performance Goal No.:			
Performance Goal Text:			
Task Description:	The experimental success in 1995 that produced the first dilute atomic Bose-Einstein condensate has led to an explosive growth of interests in this new type of degenerate quantum matter. As a testing ground for many fundamental issues in physics, in particular, in our understanding of interacting many body systems, atomic quantum gas has attracted much recent theoretical and experimental efforts. Ultimately, to grow beyond the novelty of being just another many body quantum mechanic phenomena, useful applications must be found that can take advantage of the macroscopic collective coherence of Bose condensed atoms, as well as the demonstrated abilities for the control of their interactions and the manipulation of their quantum states. At the frontier of atomic quantum gas research, a substantial topic of current interest is the creation of quantum correlated states of condensed atoms for several potential applications ranging from quantum computation to high precision quantum limited atom interferometry. In this proposal, we will investigate in detail the generation, detections		

Task Description:	and application of quantum correlated states of Bose condensed atoms. We will focus our efforts on two particular types of states that display inseparable quantum correlations: spin squeezed states and maximally entangled states. We will also study their relevance to several potential applications of trapped quantum gases. Our primary scientific objective to address the fundamental understanding of the proposed topics. We will develop new theoretical ideas and provide explicit estimate for parameters and conditions relevant to their technological applications. We expect our program will indirectly benefit experimental groups. To understand nonclassical correlation and/or entanglement properties of Bose condensed atoms, and to dvelope effective means for their generation and detection. Spin squeezing and entanglement in an ensemble of fixed numbers of atoms can be used to enhance signal to noise ratios of certain atomic population based measurements, particularly for the atomic clock/frequency measurements based on the Ramsey interference scheme.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	This study constitutes an important part of the theoretical efforts in support of improved atomic clock technology.
Task Progress:	In collaboration with the experimental group of Dr. M.S. Chapman at Georgia Tech, we have completed a theoretical study of the temperature dependent phase diagram of atomic Bose-Einstein condensation in a spin-1 gas. We have studied inseparable quantum correlation of three bosonic qubits, i.e. a system of three identical bosons in two modes. We have clarified the relationship between entanglement and spin queezing in such a system.
Bibliography Type:	Description: (Last Updated: 08/26/2021)
Articles in Peer-reviewed Journals	Zhang WX, Yi S, You L. "Bose-Einstein condensation of trapped interacting spin-1 atoms." Physical Review A. 2004 Oct;70(4):043611. https://doi.org/10.1103/PhysRevA.70.043611 , Oct-2004
Articles in Peer-reviewed Journals	Zeng B, Zhou DL, Xu Z, Sun CP, You L. "Encoding a logical qubit into physical qubits." Physical Review A. 2005 Feb;71(2):022309. https://doi.org/10.1103/PhysRevA.71.022309 , Jan-2005